Uncertainty-based multiobjective optimization of groundwater remediation design

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[1] Management of groundwater contamination often involves conflicting objectives and substantial uncertainty. The primary objective of this paper is to present a new approach, called the probabilistic multiobjective genetic algorithm (PMOGA), which incorporates uncertainty with multiobjective Pareto optimization. This type of approach is needed for water resources problems with multiple uncertain objectives; previous uncertainty-based optimization approaches consider uncertainty only in the constraints (typically constraining reliability with respect to environmental standards) or in a single objective. The optimization algorithm uses a probabilistic ranking and crowding scheme that improves decision making within the genetic algorithm by identifying a set of solutions that are Pareto optimal despite the uncertainty in the objective values. The algorithm is applied to two groundwater remediation test cases with uncertain objectives: a hypothetical case study and a field-scale pump-and-treat design problem at the Umatilla Chemical Depot situated at Hermiston, Oregon. For these case studies the primary source of uncertainty stems from uncertain aquifer hydraulic conductivity values, which affect both the remediation cost and efficiency. Results are analyzed, and the advantages of the PMOGA are discussed relative to an averaging-based multiobjective approach, a stochastic single-objective approach (similar to chance constrained optimization), and a deterministic multiobjective approach. The results demonstrate that using such an uncertainty-based multiobjective optimization scheme can give valuable information about remediation options, giving results that are not only cost effective but that also have lower uncertainty.

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1. Introduction

[2] Most real life water resources optimization problems are complex, inherently uncertain, and have multiple conflicting objectives. Groundwater remediation at a contaminated site is one such case where multiple criteria may need to be considered. Typical objectives for groundwater remediation are minimization of cleanup cost, minimization of risk posed from the contaminated site, maximization of cleanup efficiency, or minimization of cleanup time. In addition there may be other economic, regulatory, or social criteria that need to be considered when multiple stakeholders are involved in the process. The problem is further compounded when these objectives are uncertain, either because of modeling approximations or because of to the inherent stochasticity in field properties and environmental conditions. Consider, for example, the Umatilla Chemical Depot [*Minsker et al.*, 2003], situated at Hermiston, Oregon, which was used as a test bed for a demonstration and validation project conducted by the Department of Defense Environmental Security Technology Certification Program (ESTCP) on the costs and benefits of using deterministic, single-objective optimization for pump-and-treat design. The remediation objectives consisted of minimizing the cost of the remediation strategy (for groundwater contaminated with RDX and TNT) and maximizing the amount of contaminant removed from the groundwater, thereby combining two of the objectives considered separately in the ESTCP project. Ideally one would want the pump-and-treat system to reduce the contaminant concentrations down to the cleanup levels (i.e., the total mass remaining above cleanup level should be zero), but the ESTCP project [*Minsker et al.*, 2003] found that such a strategy requires an inordinately large cost to clean up small amounts of residual contamination in the groundwater. From a practical perspective, such residual contamination may best be cleaned up using cheaper long-term remediation strategies such as bioremediation or phytoremediation. In such circumstances, one needs to find the best compromise between the predicted cost and cleanup levels for different pump-and-treat configurations. In addition, the predicted cost and cleanup levels are uncertain; the predicted mass remaining at the site is uncertain as it depends on unknown field conditions (in particular hydraulic conductivity), while the (predicted) cost is also uncertain as it depends on the cleanup duration and treatment strategy. Because of these uncertainties it becomes much more complicated to assess

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the ''true'' trade-offs between solutions. For these reasons, the Umatilla site team was interested in a follow-up study to (1) identify groundwater remediation strategies that represented the optimal trade-off between multiple uncertain remediation objectives, and (2) consider the effect uncertainty has on this trade-off and the remediation strategy chosen.

[3] The goal in such multiobjective cases is to find the set of Pareto-optimal solutions, solutions such that one objective cannot be improved without worsening at least one other objective. There are three prevalent methodologies to solving such problems. The first is based on a weighting approach, wherein all of the objectives are combined using a set of weighting coefficients, thus converting the multiobjective problem into a single-objective one. Since the objectives are uncertain, ''noisy'' single-objective optimization approaches such as *Fitzpatrick and Grefenstette* [1988], *Miller and Goldberg* [1996], *Miller* [1997], and *Gopalakrishnan et al.* [2003] could be used to find the solution representing the optimal objective levels for the given weighting coefficients. It can be proved that this solution for the single objective formulation lies on the Pareto front for the multiobjective formulation [*de Weck*, 2004]. However, there are two problems with this approach: First, depending on the weighting, the single-objective solution would lie on different parts of the multiobjective Pareto front, thus the optimization would have to be run multiple times to generate the entire front; second, weighting objectives is only effective in finding Pareto-optimal solutions along the convex portion of the front, thus solutions along nonconvex portions of the front could not be found by this method [*Das and Dennis*, 1997].

[4] The second approach, often called ''goal programming,'' is to pose all but one objective as constraints and use this remaining (single) objective to optimize the problem. In such a formulation, both the objective and the constraints would be uncertain and ''chance constrained'' optimization algorithms [*Charnes and Cooper*, 1959] would need to be used to find the optimal solution for the given constraint level at a prespecified reliability level. Violations of the constraints are added to the single objective using penalty functions. Despite the fact that such an approach does find solutions in the nonconvex regions of the Pareto front, it still needs to be run multiple times with different constraint levels to identify the entire trade-off curve. Another drawback of this approach is that the user must select constraint levels through a tedious process of trial and error because the shape of the Pareto front is not known a priori, so key points along the trade-off front can be missed. Finally, the appropriate penalty function for the different constraints is not known before hand and needs further trials to be set correctly before the optimization algorithm is run.

[5] Thus any approach that is based on reformulating the multiobjective problem as a (weighted or constrained) single-objective problem only identifies one (optimal) combination of the objectives and thus needs to be run multiple times to identify the entire trade-off curve. The additional runs increase the computational cost required to generate the Pareto front, particularly for a computationally intensive application like the Umatilla case. To overcome these drawbacks a third Pareto optimization approach has been proposed to find multiple solutions along the Pareto front simultaneously without any weights or penalties. The approaches that have become increasingly popular in recent years are population-based, global optimization methods like multiobjective genetic algorithms (MOGAs), which simultaneously find multiple solutions on the Pareto front in a single run. MOGAs have been successfully applied to many water resources applications [e.g., *Cieniawski*, 1993; *Ritzel et al.*, 1994; *Halhal et al.*, 1997; *Erikson et al.*, 2002; *Reed et al.*, 2001; *Reed and Minsker*, 2004]. However, most of the previous work in the field of Pareto optimization has not considered uncertainty in the objectives. To address this gap, this paper proposes a new approach called the probabilistic multiobjective genetic algorithm (PMOGA), which simultaneously finds multiple solutions along the Pareto front while explicitly considering uncertainty in the multiple objectives during the genetic algorithm decision making process. The PMOGA is a confluence of two streams of research within genetic algorithms, noisy genetic algorithms that optimize a single objective under uncertainty [*Fitzpatrick and Grefenstette*, 1988; *Miller and Goldberg*, 1996; *Smalley et al.*, 2000; *Gopalakrishnan et al.*, 2003; *Aly and Peralta*, 1999; *Chan Hilton and Culver*, 2005] and multiobjective genetic algorithms (MOGAs) (see *Coello* [1999] for a review on multiobjective genetic algorithms) that are used to optimize multiple objectives (usually without uncertainty).

[6] The effect of uncertainty on Pareto-optimization problems is illustrated in Figure 1. As can be seen, instead of a well-defined ''Pareto front,'' under uncertainty, a range of solutions exist that can be thought to be Pareto optimal with a high probability (i.e., they have the highest probability of being Pareto optimal compared to other solutions). In Figure 1 the solutions shown by solid circles are clearly nondominated, since there are no solutions that are better in both objectives (here we assume the objectives are being minimized). The solutions shown by the open circles are clearly dominated because, even with consideration of uncertainty in objectives, they are worse than the nondominant solutions. However, the solutions shown by the hashed circles lie within the uncertainty range of the objectives, and a probabilistic decision needs to be made whether these solutions are dominated by the nondominant solutions or are in fact part of the Pareto-optimal front. Moreover, the uncertainty associated with each solution can vary depending on the decision variables associated with the solution. Note that even though Figure 1 shows the uncertainty region to be approximately equal across the Pareto front, this is not always the case in real applications such as the Umatilla case, where the uncertainty for a particular solution can be dependent on the solution itself. Thus, in addition to finding solutions with higher probability of being Pareto optimal, solutions with lower uncertainty in the objectives need to be preferred (over solutions with higher uncertainty) to ensure more reliable decision making. At this stage we could have used a constrained single objective approach, where one of the objectives is converted to an uncertain (chance) constraint that needs to be satisfied with a given level of reliability, while the other objective is used for optimization. However, the problems with applying this approach are the same as those stated for the deterministic constraint-based approach (discussed above): difficulty in setting appropriate constraint levels and penal-

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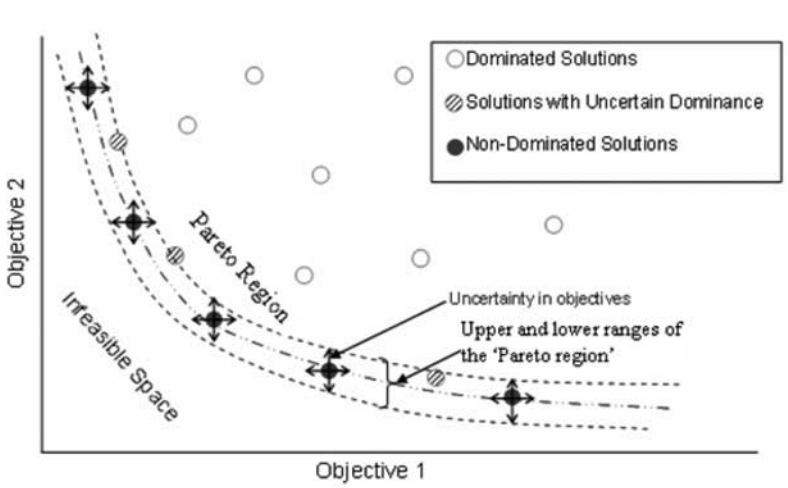


Figure 1. Uncertainty in multiobjective optimization.

ties for violations and the need for multiple runs to generate the entire Pareto front. This difficulty motivated the development of the approach given in this paper.

[7] Most previous approaches for solving water resource optimization problems with uncertainty have only considered uncertainty in the constraints or in a single objective. Apart from this study only one other group [*Beckford et al.*, 2003] has considered combining MOGAs with uncertainty. *Beckford et al.* [2003] used a single realization every generation (averaging the constraint penalty cost over generations) and used the concept of "age" in their selection procedure to find robust solutions. The difference between this study and *Beckford et al.* [2003] is that the latter considered uncertainty only in satisfying the water quality constraint and treated all other objectives deterministically (thus in essence it was considering uncertainty only for one objective). Thus, to the best of our knowledge, this work is the first to address water resources problems with uncertainty in multiple objectives within a Pareto-optimization framework.

[8] The rest of the paper is organized as follows. First two test cases, a hypothetical aquifer based on the Borden site and the field-scale application for the Umatilla Chemical Depot, are introduced. The PMOGA methodology is then described in detail in the next section. Salient design issues are addressed by demonstrating our approach on the hypothetical test case, which serves as an efficient testing ground for the PMOGA. The PMOGA is then applied to a field-scale groundwater remediation design problem at the Umatilla Chemical Depot, which was created with input from the site personnel. Finally we present our conclusions with an overview of the important features and findings of this research.

2. Test Cases

[9] The proposed methodology was developed and tested on a hypothetical groundwater remediation problem, which allowed us to rapidly test different approaches and eliminate those that did not show much promise. The best method was then implemented for the Umatilla test case to see how the algorithm behaved in a real life scenario. This section describes both test cases.

2.1. Hypothetical Groundwater Remediation Test Case

[10] This risk-based remediation design case study has been developed using data derived from the Borden site as described by *Smalley et al.* [2000]. The confined, heterogeneous, isotropic and steady state aquifer has the dimensions $480 \times 240 \times 20$ meters and was modeled using a coarse grid of 16 by 8 elements. This coarse grid was derived from a finer mesh that was used to generate conditional hydraulic conductivity realizations as described by *Feyen et al.* [2001]. BTEX (benzene, toluene, ethylbenzene, xylene) with an initial peak concentration of 133 mg/L was assumed to be present up-gradient at the site (see Figure 2). Pump-and-treat technology was used for the remediation process with an ex situ air stripping technology to treat the extracted water. A maximum of three remediation wells can be selected for extraction/injection (maximum pumping capacity is set at 250 m^3/d) at the 58 candidate locations (shown as black dots in the center of the grid in Figure 2). A risk-based remediation plan was developed for the case study using the risk management model developed by *Smalley et al.* [2000] and *Gopalakrishnan et al.* [2003]. The management model combines a genetic algorithm with a fate and transport simulation model and a risk assessment module to identify promising remediation designs.

[11] Two objectives are considered for this case. The first objective is to minimize the total treatment cost (C_{TOT}) function, which consists of remediation well costs (C_{REM}), monitoring costs (C_{MON}) for the 15 monitoring wells (shown by open squares in Figure 2), and ex situ treatment costs (C_{SYST}), as follows:

$$C_{TOT} = \sum_t \sum_i C_{REM}(x_i, q_i, sq_i, t) + C_{MON}(t) + C_{SYST}(q_i, t) \quad (1)$$

where, x_i is the location of well i, q_i is the pumping rate decision variable for well i, sq_i is the flag that determines whether well i is to be used for extraction or injection, and t

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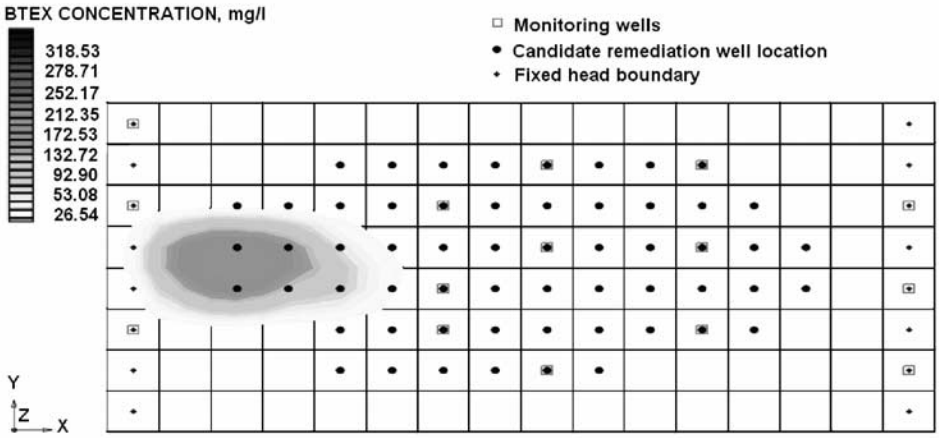


Figure 2. BTEX plume on coarse grid size for the hypothetical case study.

is the length of remediation period. The detailed equations for C_{TOT} are extensive and are documented by *Smalley et al.* [2000] and *Gopalakrishnan et al.* [2003]. It is important to note that the ex situ treatment costs (C_{SYST}) depend on the volume of water withdrawn from the pumping wells and the BTEX concentration in the treated water. Thus this part of the remediation cost depends on the prevailing hydraulic conditions and hydrogeology of the site and is uncertain because of uncertain aquifer parameters for the groundwater model.

[12] The second objective is to minimize the total human health risk, $Risk^{Total}$:

$$Risk^{Total} = \sum_t \sum_k Risk^{w}_{t,k}(x_i, q_i, sq_i, t) + Risk^{shw}_{t,k}(x_i, q_i, sq_i, t) + Risk^{nc}_{t,k}(x_i, q_i, sq_i, t) \quad (2)$$

where, $Risk^{w}_{t,k}$, $Risk^{shw}_{t,k}$, $Risk^{nc}_{t,k}$, are the cancer risks at time t and exposure location k due to ingestion of contaminated drinking water, inhalation of volatiles from contaminated water due to showering, and inhalation of volatiles from contaminated water due to other nonconsumptive uses, respectively. These risks depend on the contaminant concentrations, which are in turn dependent on the well locations (x_i), pumping rates (q_i), and the length of the remediation period (t). To calculate these risks, MODFLOW [*McDonald and Harbaugh*, 1988] and RT3D [*Clement*, 1997] are used to simulate contaminant concentration in the source area for each candidate solution. The concentrations of the plume estimated by the numerical model are then used to predict the above human health risks using an analytical stochastic risk assessment model developed by *Smalley et al.* [2000] at an assumed exposure location 200 m offsite (see *Smalley et al.* [2000] for details on how these terms are calculated). The uncertainty in the risk estimate is due to the uncertainty in the predicted contaminant concentrations (which in turn are related to uncertainty in aquifer parameters) as well as the uncertainty in the human exposure parameters used to relate contaminant concentrations to cancer risks on consumption. It is important to note that, unlike *Gopalakrishnan et al.* [2003], risk is considered to be an objective here and not a constraint, precluding the application of approaches based on chance-constraint optimization.

[13] The hard constraints for this problem include pumping rate limits ($q_{min,i} \leq q_i \leq q_{max,i}$, where $q_{min,i}$ and $q_{max,i}$ are the lower and upper limits of the pumping rates for well i), and hydraulic drawdown limits ($h_{min,i} < h_i < h_{max,i}$, where and $h_{min,i}$ and $h_{max,i}$ are the lower and upper limits of the drawdowns for well i). The decision variables are encoded into a binary string with 48 bits. A population size of 300 and 100 generations were used for the GA runs. The mutation rate was set at 0.0033 (inverse of the population size) and the crossover rate was kept at 0.80. The GA parameters were set using guidelines outlined by *Reed et al.* [2003].

[14] To consider uncertainty at this site, multiple parameter sets are defined, with each set consisting of a single sample drawn randomly from an ensemble of 500 hydraulic conductivity realizations and from each of nine variable human exposure parameter (used in the risk assessment model) distributions. The hydraulic conductivity realizations are generated using geostatistical conditional simulations (for details see *Smalley et al.* [2000]). It is worth mentioning here that the uncertainty for a particular remediation design solution depends not only on the conductivity scenario chosen but on the decision variables, location and pumping rates for wells, for that solution. This has important implications on the design of the uncertainty-based optimization algorithm to be discussed in subsequent sections.

2.2. Umatilla Chemical Depot Field Application

[15] The Umatilla Chemical Depot at Hermiston, Oregon has been chosen as a field-scale test case. This 19,728-acre military reservation was established in 1941 and operated an onsite explosives washout plant from the 1950s until 1965. During this time around 85 million gallons of wash water was discharged into two unlined lagoons at the site. The two major chemicals released were 2,4,6-tri-nitro toluene (TNT) and hexahydro-1,3,5-trinitro-1,3,5-triazine (commonly known as RDX). Although regular dredging of the lagoons was undertaken, explosives contained in the wash water migrated into the soil and groundwater at the site.

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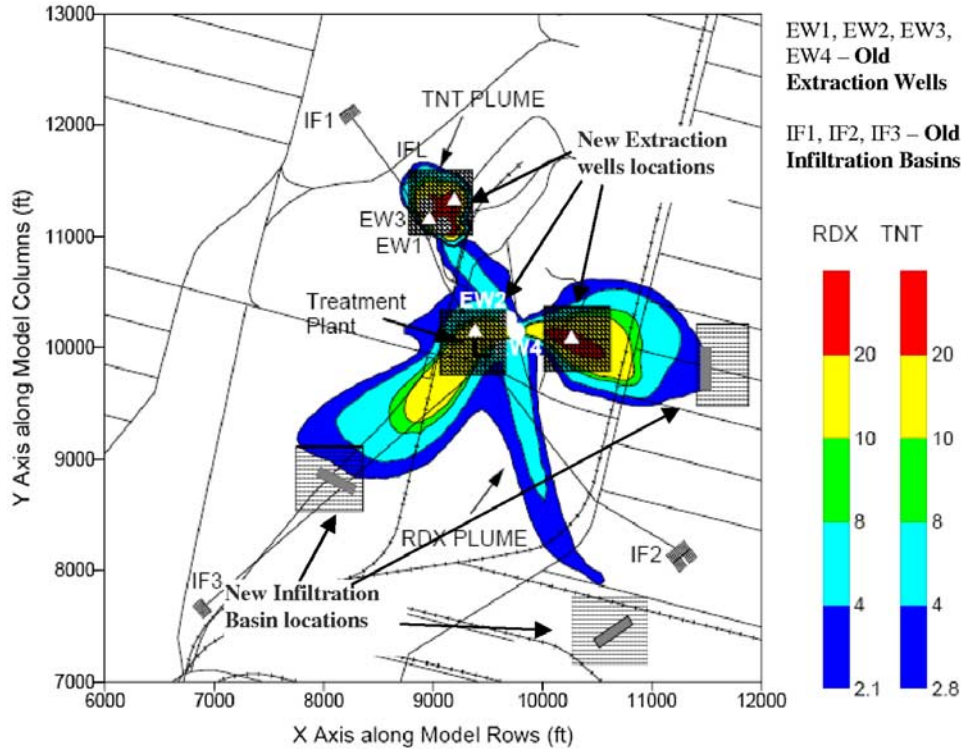


Figure 3. Pump-and-treat system (as of October 2000). The shaded areas represent zones for potential extraction wells (EW) and infiltration basins (IF). Figure is taken from *Minsker et al.* [2003].

[16] To remediate this contamination, a pump and treat system (see Figure 3) consisting of 4 extraction wells (one of which is not currently operational) and 3 infiltration basins is operating at the site. The cleanup levels for RDX and TNT have been set at 2.1 μg/l and 2.8 μg/l, respectively.

[17] This test case was first examined by the Department of Defense Environmental Security Technology Certification Program (ESTCP) team to demonstrate and validate transport optimization methodologies [*Minsker et al.*, 2003]. Two of the objectives considered (separately as single objectives) by the ESTCP team were minimizing total treatment cost and minimizing total mass remaining. Because of the dependence of the optimal pumping rates on the hydraulic conductivity, there is considerable uncertainty in both these objectives that was not considered in the ESTCP project. The formulation for this study was created with input from Umatilla site personnel who were interested in learning more about the tradeoffs among these uncertain objectives.

2.2.1. Formulation

[18] The first objective to be minimized is the total cost for the pump-and-treat system. The second objective is the total mass of contaminant above the cleanup concentration levels remaining in the simulated aquifer. It is important to note that even though there are existing standards for the contaminant concentrations there are no such constraints on the total mass remaining at the site. Because of this reason it is difficult (and perhaps unreasonable) to pose the remaining mass objective as a constraint, again precluding constraint-based optimization methods. In this formulation, we allow a maximum of 4 new wells and 3 new recharge basins to be installed, in addition to choosing any of the existing 4 wells (the currently nonoperational well is also considered) and 3 basins. The total cost is given by

$$\begin{aligned} Cost^{Total} = \sum_t \sum_i & CCW\left(t, u_{i,nw}\right) + CCB\left(t, u_{i,nrb}\right) \\ & + CCG\left(t, q_{i,nw}, q_{i,ow}, u_{i,nw}, u_{i,ow}\right) \\ & + FCL(t) + FCE(t) \\ & + VCE\left(t, q_{i,nw}, q_{i,ow}, u_{i,nw}, u_{i,ow}\right) \\ & + VCG\left(t, x_{i,nw}, q_{i,nw}, q_{i,ow}, u_{i,nw}, u_{i,ow}\right) \\ & + VCS\left(t, x_{i,nw}, q_{i,nw}, q_{i,ow}, u_{i,nw}, u_{i,ow}\right) \\ & + VCS\left(t, x_{i,nw}, q_{i,nw}, q_{i,ow}, u_{i,nw}, u_{i,ow}\right) \end{aligned} \quad (3)$$

where t represents the remediation years, $x_{i,nw}$ is a decision variable for the location of the ith new well/infiltration basin, $q_{i,nw}$ is a decision variable for the pumping rate at the ith new well/infiltration basins, $q_{i,ow}$ is a decision variable for the pumping rate at the ith existing well, $u_{i,nw}$ is a binary decision variable for installing/using the ith new well/infiltration basin, $u_{i,ow}$ is a binary decision variable for installing the ith old well, $u_{i,nrb}$ is a binary decision variable for installing the ith new infiltration basin, CCW is the capital cost of new wells, CCB is the capital cost of new infiltration basins, CCG is the capital cost of a new granulated activated carbon (GAC) treatment unit, FCL is the fixed cost of labor, FCE is the fixed cost of electricity, VCE is the variable electrical cost of operating wells, VCG is the variable cost of changing GAC units, and VCS is the variable cost of sampling.

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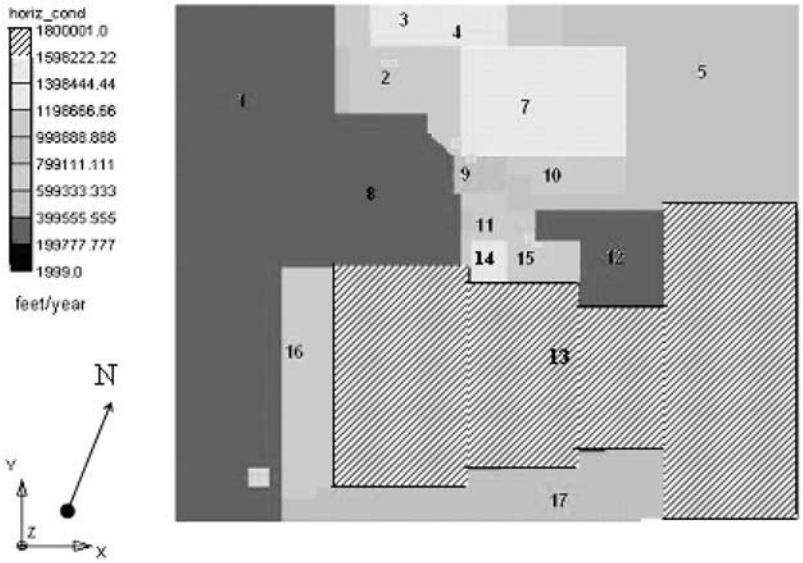


Figure 4. Hydraulic conductivity zones used in the Umatilla model.

[19] The objective to minimize mass remaining above cleanup level after the remediation period is defined by

$$Total\ Mass = \sum_{c} \sum_{i} \mathrm{Max}\left(0, Conc_{c,i} - Conc_{c}^{\mathrm{Max}}\right) * Vol_{i} \quad (4)$$

where $Conc_{c,i}$ is the concentration of contaminant c in cell i (of the modeled aquifer), Vol_i is the volume of the cell, and $Conc_c^{\mathrm{Max}}$ is the cleanup level as given by the maximum allowable concentration for contaminant *c*. MODFLOW and MT3DMS [*Zheng and Wang*, 1999] were used to predict the contaminant concentrations at every point in the modeled grids. The Max( ) function imposes the condition that the mass of the particular contaminant in a cell is only considered if the predicted concentration is above the cleanup level. Further details on the components of these objective functions are given by *Minsker et al.* [2003].

[20] Similar to *Zheng and Wang* [2002] the new wells and basins were constrained to lie within selected zones (shown by the shaded rectangles in Figure 3) to reduce computational effort. Each zone had a certain number of grid cells that could be chosen as well locations. Five of these zones contained 32 possible locations (represented by 5 bits in the binary GA) and two contained 53 well locations (represented by 6 bits). The 8 pumping rates (for the 4 new and the 4 existing wells) were discretized to 4 levels (represented by 2 bits), representing percentages of the highest allowable pumping rate in a particular zone. Thus the pumping locations contributed 37 bits, the installation flags 14 bits, and the pumping rates 16 bits to the entire encoded chromosome length of 67 bits. A population size of 250, 134 generations, mutation rate of 0.004, and crossover rate of 0.8 were used for all the runs, based on the guidelines of *Reed et al.* [2003].

[21] The constraints for the formulation are: the current GAC treatment capacity of 1300 gallons per minute (gpm) must not be exceeded; pumping must be limited to 400 gpm in a low-conductivity zone (top part of Figure 4) and 1000 gpm in the high-conductivity zone (bottom part of Figure 4) with 10% pumping downtime; concentrations cannot exceed the cleanup levels outside a specified area; and mass balance is imposed by keeping total extraction equal to total injection. For more details on the existing design or the objective function and constraints developed in the ESTCP project, see *U.S. Army Corps of Engineers* (*USACE*) [1996, 2000].

[22] In the ESTCP project, *Zheng and Wang* [2002] and *Peralta* [2002] identified candidate design solutions for the location and pumping rates of the wells and basins for three single-objective formulations. The first two formulations involved minimizing the total remediation cost until cleanup levels were achieved and the third formulation involved minimizing the total mass of pollutants remaining in the groundwater after 20 years. A major finding in their solutions was that the most cost-effective approach to remediation would be to shut down the pumping wells in the RDX plume and focus pumping in the TNT plume (see Figure 3). This approach predicted cleanup in 4 years at a lower cost than the existing solution. The feasibility of this approach is dependent on the existence of relatively high estimated hydraulic conductivities at the southern end of the site (seen as the hatched area in Figure 4), which allow the RDX plume to easily be drawn north to wells in the TNT plume. Recently, *Peralta and Kalwij* [2003] applied a robust single-objective genetic algorithm (that takes into

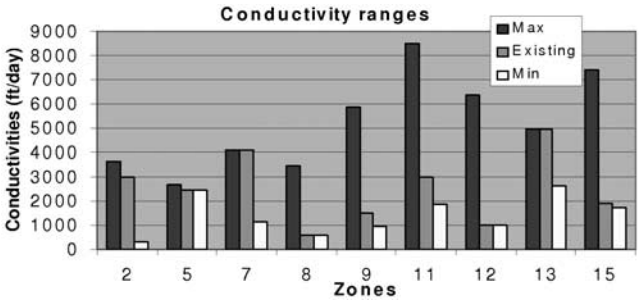


Figure 5. Conductivity ranges from pumping tests (only zones with sampling results are shown).

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account the uncertainty of the solutions) to the Umatilla Chemical Depot site to improve reliability of the Umatilla cleanup solutions. They noted that solutions with similar costs can have different reliabilities with respect to the hydraulic conductivity values. This paper follows up on their study by using PMOGA to identify more reliable multiobjective designs.

2.2.2. Characterization of Uncertainty

[23] For the Umatilla site we focus on the uncertainty in the hydraulic conductivities used for the aquifer zones. We used a simple methodology to characterize the uncertainty with the limited amount of pumping data available for the Umatilla site. The data available are the pumping test results conducted by the USACE for the 47 monitoring wells to identify hydraulic conductivity values (shown in Figure 5) for each of the zones given in Figure 4 [*USACE*, 1996, 2000]. Of the 17 zones shown in Figure 4, pumping test data were available for only 9 zones. The USACE set the boundaries and values for each zone by calibrating the model to observations of the annual average groundwater elevations for the area using professional expertise and judgment. These conductivity values (shown as the ''existing case'' in Figure 5) were then used for subsequent simulations of the fate and transport of the contaminant species by the USACE and in the ESTCP project.

[24] To model the uncertainty in these conductivity values, we generated realizations with different zone values. Because of the lack of extensive data (only 47 monitoring wells in 9 out of 17 zones), more sophisticated geostatistical approaches could not be used for this site. Instead, a simpler approach was taken in which realizations were generated by sampling from the ranges for each zone established from the pumping tests shown in Figure 5. A triangular distribution was used for each zone with the existing values for the conductivities (which give a close fit between the predicted and measured hydraulic heads) taken as the mode (apex) and the lower and upper bounds given by the minimum and maximum conductivities from the pumping tests. A total of 100 realizations were created in this manner. Sampling was carried out from these realizations of the hydraulic head scenarios using efficient Latin hypercube sampling (LHS) as described by *McKay et al.* [1979]. Recently *Zhang and Pinder* [2000] demonstrated the utility of this technique for generating random hydraulic conductivity fields, showing that the method gave better results for the same computational cost as Monte Carlo sampling. First moments of the absolute differences in simulated head values (which we refer to as the ''difference moments'', please see *Singh and Minsker* [2003] for details) relative to the current model (which the USACE considered to be a good fit to observed heads) give a measure of how different a particular realization is from the current conductivity scenario with respect to the predicted hydraulic heads. The difference moments were used for stratifying the LHS sampling as detailed by *Singh et al.* [2003]. Four of the hundred realizations had very high difference moments, indicating that these realizations were unrealistic and led to predicted heads significantly different from the existing heads. On the basis of the recommendations of the USACE these realizations were discarded from the pool and the remaining 96 were used for sampling. Using this methodology for efficient sampling, each candidate design could be tested over various conductivity realizations within the probabilistic multiobjective genetic algorithm.

3. Methodology

[25] Figure 6 gives an overview of the PMOGA methodology. PMOGA begins like any genetic algorithm [*Goldberg*, 1989] randomly initializing a population of potential solutions (''individuals'') to the remediation design problem. Each individual consists of a binary representation of the decision variables (in the cases studied in this paper, the locations and pumping rates of the wells, as well as infiltration basins for Umatilla). The PMOGA consists of two components: the probabilistic part and the ''multiobjective'' part. The probabilistic component of PMOGA handles uncertainty by sampling each individual multiple times with different realizations of the uncertain parameters (in this case conductivity scenarios) and evaluating the objective functions for each realization. This process is similar to other noisy GAs [*Fitzpatrick and Grefenstette*, 1988; *Miller and Goldberg*, 1996; *Gopalakrishnan et al.*, 2003]. We have introduced an additional step in the sampling phase that stores evaluated solutions in an archive for future use. The sampling phase is discussed in more detail in section 3.1. Once the sampling has been completed, PMOGA calculates sample statistics for each objective and begins the multiobjective operations. For this paper, we created the PMOGA multiobjective operators by adapting the nondominated sorting genetic algorithm (NSGA-II) proposed by *Deb et al.* [2000], which has been shown to be an efficient and robust multiobjective algorithm. The NSGA-II uses selection operators based on Pareto ranking and crowding distances of the solutions. To modify these operators to account for uncertainty in the objective values, we propose the probabilistic ranking and probabilistic crowding operators that are discussed in detail in sections 3.2 and 3.3, respectively. Once the probabilistic ranking and distance calculations are completed, these are used for multiobjective selection to identify individuals for mating to create the next generation. PMOGA uses elitist selection to preserve the best solutions found, which is similar to the standard NSGA-II except that it uses probabilistic ranking and crowding distance, as discussed in more detail in section 3.4. After selection, PMOGA uses the standard GA operators of crossover and mutation [*Goldberg*, 1989]. This process is repeated until a convergence criterion is met (in this case the maximum number of generations).

[26] Once the Pareto front has converged, it gives the trade-offs between different objectives in the face of uncertainty, where the probabilistic ranking and crowding operations ensure that each solution found is the most robust solution for a given value of each objective. At this stage, depending on what costs and risk levels the decision maker is interested in, portions of the Pareto front (or the entire front) can be selected for more extensive off-line Monte Carlo analysis to further test the reliability of the solutions as demonstrated in the results section.

3.1. Sampling for PMOGA

[27] An issue that arises when considering uncertainty is how to best assess and evaluate the true uncertainty in the problem. Chance constrained approaches typically model the uncertainty analytically by assuming a known distribu-

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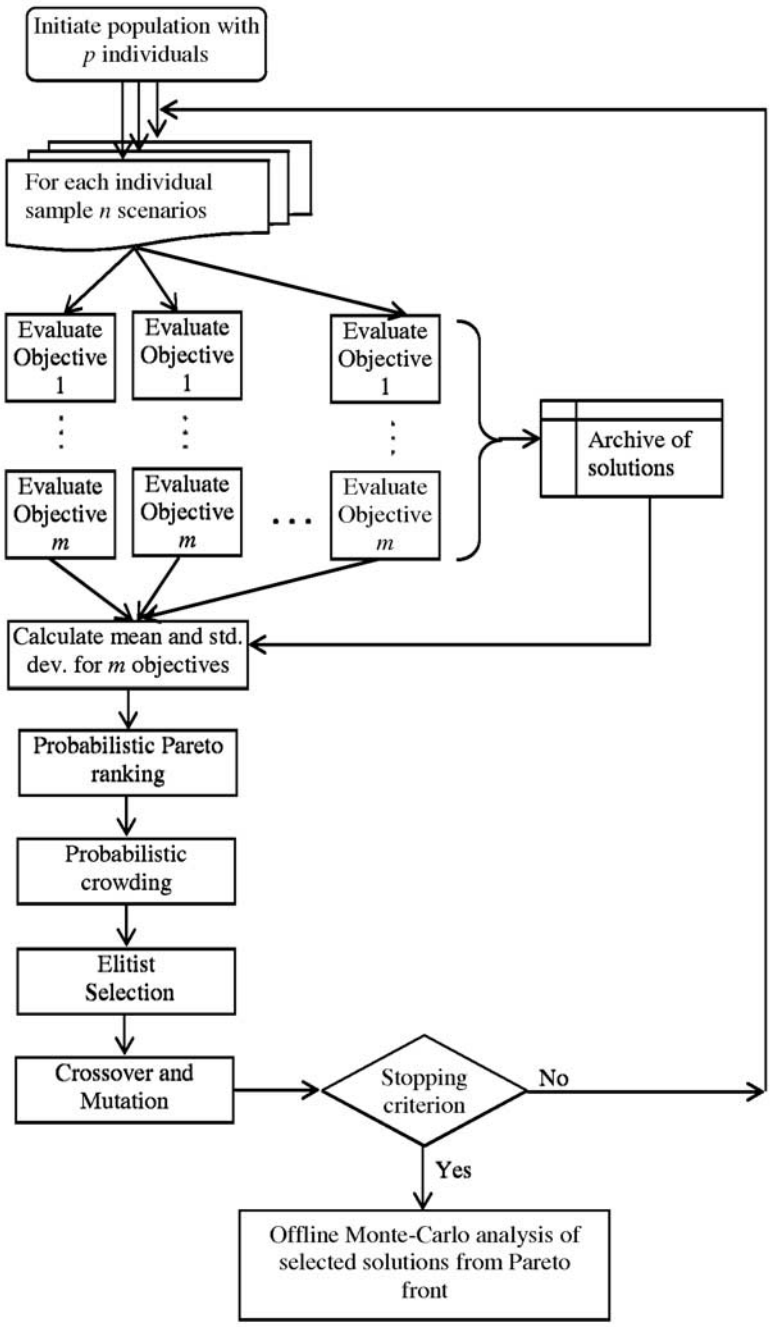


Figure 6. Overview of the probabilistic multiobjective GA (PMOGA).

tion for the uncertainty and converting the stochastic constraints to deterministic equivalents [examples are *Wagner and Gorelick*, 1987; *Tiedeman and Gorelick*, 1993; *Chan*, 1994; *Guan and Aral*, 2004]. An alternative nonparametric approach is to generate multiple equally likely scenarios and use these to estimate the uncertainty using Monte Carlo-based sampling (examples include *Wagner and Gorelick* [1989], *Chan* [1993], and *Morgan et al.* [1993]). Since the second approach is nonparametric and does not require any assumptions about the distribution of the uncertainty in the objectives, it is more appropriate for the Umatilla application and is used in the current research.

[28] The sampling phase of PMOGA consists of evaluating the objectives for multiple realizations of the uncertain variables, which in this case involves running the MODFLOW and MT3DMS model for each solution and evaluating the objective functions for multiple conductivity realizations. Once the objective function values for all realizations are evaluated, sample statistics (mean, standard deviation, and number of samples) are calculated for each objective. These statistics are then used for the selection phase of the GA.

[29] To improve reliability of the estimated objective functions, all of the sample statistics for a particular individual are stored in an archive and used for subsequent

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statistical calculations. In subsequent generations, whenever the sample statistics are being calculated for any individual, this archive is checked to identify any duplicates in storage. If duplicates are found, then the sample statistics are updated to consider all present and past samples in the archive. We call this process extended sampling. Thus, as a particular individual survives over many generations, its effective sample size increases and the objective function statistics become more reliable. It is interesting to note that this is in some ways similar to *Beckford et al.* [2003] sampling technique across generations, the difference being that they used only a single realization every generation. The sample statistics are used in the probabilistic ranking and crowding distance calculations, which form the core of the PMOGA.

[30] Most noisy GAs [e.g., *Gopalakrishnan et al.*, 2003; *Miller*, 1997; *Beckford et al.*, 2003] use only the average value (over multiple samples) of the objective for selection. This average value increasingly tends to the actual mean of the noise distribution with increased sampling (by the central limit theorem). *Miller and Goldberg* [1996] were among the first to analyze and suggest adequate sample size and populations for single-objective noisy genetic algorithms, and *Gopalakrishnan et al.* [2003] demonstrated their approach on a single-objective groundwater remediation design problem. They found that highly reliable solutions could be obtained by simply averaging the objective value over a small number (as few as 5) of samples. Small sampling sizes work well for single-objective problems because, as the population converges, more and more identical solutions are available in the population, increasing the effective sampling size. Only solutions that perform well in all instances can take over the population, leading to reliably optimal results. However, as we pointed out earlier, multiobjective GAs do not converge to one solution thus simple averaging alone did not work in this case [*Singh et al.*, 2003], thereby motivating the probabilistic approaches described in the next sections.

[31] Note that the focus of this research is to develop a multiobjective optimization methodology to handle uncertainty (in whatever way one chooses to evaluate it). We have thus not considered alternative ways to represent the uncertainty for the chosen applications. However, the PMOGA can be applied, without any loss of generality, using any other uncertainty-modeling approach.

3.2. Probabilistic Ranking in PMOGA

[32] Multiobjective GAs (MOGAs) find the set of solutions that composes the nondominated (or Pareto) front for the problem, for which no improvements can be made in one objective without worsening at least one other objective. The NSGA-II identifies Pareto-optimal solutions using a layerwise ranking scheme. Each individual is compared to all others in the population and those that are strictly nondominated are marked as the locally Pareto (or rank 1) solutions. These are then removed from the list of considered solutions; rank 2 solutions are defined as those that are strictly nondominant with respect to the remaining solutions and so on until all solutions in the population are ranked. Once all the individuals in a population have a rank, the objective of the NSGA-II is to minimize the rank.

[33] To modify this approach to consider uncertainty, we propose a new probabilistic ranking scheme that considers both the mean and the variance of the sampled objective values to decide whether a particular solution is nondominated or not. Referring back to Figure 1, this is akin to deciding whether the solutions shown by the hashed circles are part of the Pareto band or not. This ranking scheme is an adaptation of *Hughes* [2001] selection scheme, in which a probabilistic model is used to test whether one GA solution is better than the other. In his method *Hughes* [2001] assumes that the true mean and standard deviations are normally distributed and are either known or can be estimated accurately without bias. This is usually not the case with real world optimization applications, where the uncertainty is unknown and there are severe restrictions in the number of samples that can be taken, leading to biased estimates of the mean and the standard deviation. The ranking scheme we propose uses the Student t distribution for the probabilistic selection and can thus better handle the small number of samples in our case. The t test also takes into consideration the differences in variance between two competing individuals. Thus our probabilistic ranking scheme incorporates not only the estimated mean and variance of the fitness of individuals, but also the reliability in these estimates.

[34] The first step in probabilistic ranking is to compare the dominance of two solutions X and Y with p objectives $(x_1, x_2 \ldots x_p)$ and $(y_1, y_2, \ldots y_p)$. Because of extended sampling the solutions may have different effective sample sizes (depending on how many instances of the solution have been stored in the archive from previous generations). Assuming that the sample means for X and Y are independent, the difference between the measured mean objectives $\bar{x}_i - \bar{y}_i$ has a Student t distribution given by

$$T_i = \frac{(\bar{x}_i - \bar{y}_i) - \left(\mu_{x,i} - \mu_{y,i}\right)}{\sqrt{\frac{S_{x,i}^2}{n} + \frac{S_{y,i}^2}{m}}} \quad i = 1, 2, \ldots p \tag{5}$$

with $\min(n - 1, m - 1)$ degrees of freedom, where, $\bar{x}_i$, $\bar{y}_i$ and $S_{x,i}$, $S_{y,i}$ are the sample means and standard deviations estimated with n and m effective sample sizes (that include stored solutions from the archive), and $\mu_{x,i}$ and $\mu_{y,i}$ are the true means for solutions X and Y, respectively [*Triola*, 2002]. In other words the random variable given by $\bar{x}_i - \bar{y}_i$ has mean $\mu_{x,i} - \mu_{y,i}$ and standard deviation $R_{xy,i}$ given by the denominator of equation (5). Given this distribution, we can obtain the confidence bounds for the difference between the measured mean values:

$$P\left[(\bar{x}_i - \bar{y}_i) - bR_{xy,i} < \mu_{x,i} - \mu_{y,i} < (\bar{x}_i - \bar{y}_i) + bR_{xy,i}\right] = \alpha \tag{6}$$

where α is the percentage confidence level, b is found from t distribution tables such that $P(-b < t < b) = \alpha$ with degrees of freedom $n + m - 2$, and $R_{xy,i}$ is the denominator of (5).

[35] Equation (6) then becomes a significance test for the null hypothesis:

$$H_i : \mu_{x,i} = \mu_{y,i}. \tag{7}$$

If both the lower and upper ranges given by (6) on $\mu_{x,i} - \mu_{y,i}$ are negative then we can say with α level of confidence

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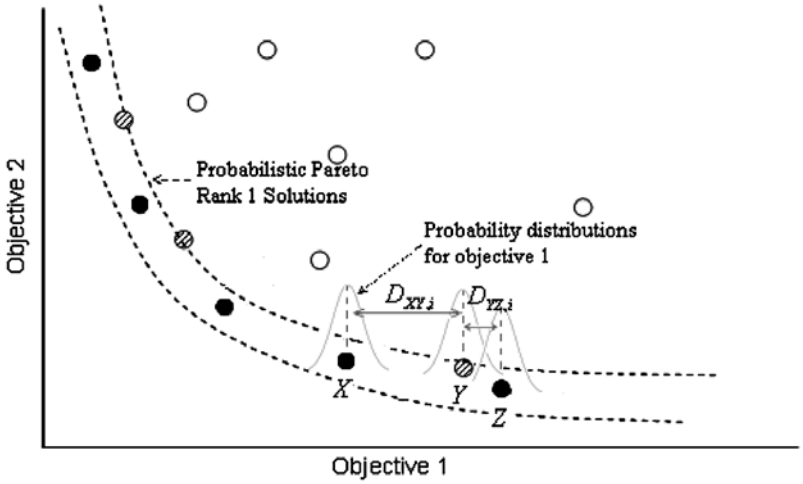


Figure 7. Probabilistic crowding distance calculation.

that $\mu_{x,i}$ is less than $\mu_{y,i}$, if they are both positive then $\mu_{x,i}$ is more than $\mu_{y,i}$ with the given confidence. In both cases the null hypothesis is rejected and we can decide statistically which of the two values is less than the other. If the range is between the upper and lower confidence bounds then we cannot reject the null hypothesis (equation (7)) and must conclude that x_i and y_i are statistically similar. This hypothesis test can be carried out for each objective. For a minimization problem, X dominates Y if $\mu_{x,i} \leq \mu_{y,i}\ \forall i$, i.e., if each objective value for X is less than or equal to the corresponding objective value for Y. This t test is used to construct "probabilistic" ranks for the NSGA-II algorithm. Given all individuals in the population, each pair is tested to see which solutions are statistically dominant. The set of individuals that are not dominated by any other solution with the given level of significance are given rank 1, the set dominated by rank 1 and dominating all others are given rank 2, and so on until all solutions are probabilistically ranked.

3.3. Probabilistic Crowding in PMOGA

[36] The NSGA-II uses "crowding distance" to preserve diversity along the Pareto front, ensuring that the final population provides good representation of the entire Pareto front. The crowding distance is a measure of a solution's uniqueness and is defined as the cumulative distance (in objective or decision space) between its neighbors summed over all objectives. During the selection process the NSGA-II first uses the rank of the solution to select one individual over another. For solutions with the same rank, the solution with the larger crowding distance is preferred.

[37] To modify the crowding operation to consider uncertainty, we introduce a probabilistic crowding approach. Similar to the NSGA-II, the probabilistic crowding distance for a particular individual is based on the (average) distance between a solution and its closest neighbors in the same Pareto front. The PMOGA calculates the probabilistic distance as the sum of the distance from a particular solution to its adjacent neighbors (note that for the NSGA-II this distance is calculated directly between the two adjacent neighbors, while for the PMOGA the distance is calculated for each neighbor from the given solution and then added to give the total crowding distance).

[38] The schematic for calculating the probabilistic crowding distance is given in Figure 7. To calculate the probabilistic crowding distance for a particular individual Y, only the solutions within the same probabilistic rank are considered. All of the solutions are then sorted on the basis of each objective. The two adjacent neighbors to individual Y are given as individuals X and Z, respectively (see Figure 7). To calculate the crowding distance between individuals X and Y, we assume that the difference between the average value of the ith objectives is a random variable with mean and variance $\boldsymbol{R}^2_{xy,i}$ (denominator from equation (5)). As before (equation (7)) the null hypothesis H_i: $\mu_{x,i} = \mu_{y,i}$ is tested using the t test given in equation (6). If the null hypothesis cannot be rejected; that is, if $\mu_{x,i} = \mu_{y,i}$ then the expected distance is considered to be zero and $D_{XY,i}$ is made equal to zero. On the other hand, if the null hypothesis is rejected; that is, if $\mu_{x,i} \neq \mu_{y,i}$ then the expected crowding distance between two individuals X and Y for the ith objective, is given by

$$D_{XY,i} = \bar{x}_i - \bar{y}_i \tag{8}$$

where $\bar{x}_i$ and $\bar{y}_i$ are the sample means (for the ith objective of the two solutions X and Y).

[39] This calculation is repeated again for individuals Y and Z. The crowding distance (corresponding for the ith objective) for individual Y is then given by

$$D_{Y,i} = D_{XY,i} + D_{YZ,i} \tag{9}$$

Finally this process is repeated for all of the p objectives and the total crowding distance for individual Y is given as the sum of the crowding distance for all objectives:

$$D_Y = \sum_{i=1}^{p} D_{Y,i} \tag{10}$$

Equation (10) can then be used as a measure for the crowding distance calculation in the PMOGA.

[40] Recall that during the selection process solutions with the larger crowding distance are favored. The proba-

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bilistic crowding distance as given by equation (10) is not only a function of the sampled means, but also the denominator in equation (5) given by $R_{xy,i}$. $R_{xy,i}$ in turn is dependent on the sampled variances $S_{x,i}$, $S_{y,i}$ and the number of times the solutions have been sampled (n and m in equation (5)). Thus the lower the sample variance and the more times each solution has survived over the generations, the lower $R^2_{xy,i}$ is going to be. Solutions with lower $R^2_{xy,i}$ would consequently be more likely to reject the null hypothesis $\mu_{x,i} = \mu_{y,i}$ (used in the calculation of crowding distance) and thus have higher crowding distance (since the $D_{XY,i}$ terms in equation (9) would be nonzero). On the other hand solutions with higher $R^2_{xy,i}$ would not be able to reject the null hypothesis and would have lower crowding distance (since $D_{XY,i}$ would be zero for such cases). In other words, larger variances and lower number of samples lead to the neighboring individuals having expected means closer to each other, which in turn leads to lower crowding distance and consequently lower probability of being selected. In this way the added advantage of this probabilistic distance scheme is that it not only favors solutions in sparse regions of the Pareto front (thereby maintaining diversity along the front), but also explicitly favors solutions with lower uncertainty, leading to more reliable results.

3.4. Elitist Selection in PMOGA

[41] To ensure that good solutions found in early generations are not lost subsequently the NSGA-II uses an "elitist" selection scheme. In this scheme the entire parent population is combined with the child population and ranking and crowding are carried out for this combined population. The top half (as assessed by the rank and the crowding distance) of individuals from this combined population are then selected and put into the subsequent generation. The PMOGA uses the same elitist scheme as the standard NSGA-II except that it uses the probabilistic rank and the probabilistic crowding distance for the selection process. When selecting the top half of the combined population, first preference is given to the rank. Ties in ranks are resolved on the basis of the probabilistic crowding distance, with individuals having higher crowding distance being preferred. Thus, by using this elitist selection scheme in conjunction with probabilistic ranking and crowding we accomplish three goals. First, solutions with low objective values are conserved (because of elitist selection). Secondly, solutions with more reliable objective value estimates (i.e., solutions with higher effective sample sizes) are preferred (because of the t test used in the ranking process). Finally, unique solutions with low variance are considered better than solutions with high variance (due to probabilistic crowding).

4. Results

[42] When testing the algorithms we first ran the PMOGA with a small sample size. To test the reliability of the results we then conducted extensive Monte Carlo simulations using the entire sample set (500 samples for the hypothetical case and 96 samples for the Umatilla case) for each solution on the final Pareto frontier. This gave us a better estimate of the "true" average as well as upper and lower confidence intervals. Upon establishing such bounds we can then compare the small sample statistics of the solutions on the Pareto frontier with the true statistics as obtained by the extensive sampling. This allows us to see where the solutions obtained from the GA (with limited sampling) lie in relation to the upper and lower confidence regions. The closer the GA solutions are to the true performance (in terms of the average obtained from full Monte Carlo analysis) the more reliable they are for decision making. Solutions with this property can be thought of as solutions with low bias. Moreover, in problems such as ours where the uncertainty is dependent on the characteristics of the individual (unlike a constant uncertainty factor due to, say, measurement error) it is also desirable that the final solutions have low variability, since less variability corresponds to higher reliability of solutions when implemented in practice. Before we discuss the results of the PMOGA on the Umatilla case, we first compare the PMOGA results for the hypothetical aquifer with the noisy single-objective approach (same as *Gopalakrishnan et al.* [2003]). This serves as a QA/QC test for the PMOGA as the single-objective results should be part of the multiobjective trade-off curve if the PMOGA has found the optimal Pareto front. Secondly we compare the PMOGA on the hypothetical aquifer with a simple averaging approach to show the advantages of using the probabilistic methodology proposed here. Finally the results of the PMOGA applied to the Umatilla case are discussed in section 4.2.

4.1. Results for the Hypothetical Aquifer

[43] To test the effectiveness of the probabilistic selection and crowding scheme used in PMOGA, we first ran the NSGA-II with a simple averaging approach (i.e., using PMOGA without probabilistic selection and crowding, albeit with extended averaging). This is similar to the approach used by *Gopalakrishnan et al.* [2003] for the single objective case with uncertainty. For each individual the average objective values are estimated from the sampling and these average values are then used within a standard multiobjective GA for ranking and crowding distance calculation. The results for the averaging approach for the hypothetical groundwater problem are shown in Figure 8. Figure 8 shows the portion of the cost versus risk trade-off curve corresponding to low risk levels (less than $10^{-3.5}$). The cost function is discontinuous because it depends on the capital cost for the well installation, which is increased for every additional well (in this case the discontinuity corresponds to a change from 2 to 3 wells) installed. In Figure 8, the triangles show the final population from NSGA-II when only a sampling size of 5 was used, while the crosses, blocks, and diamonds show the average, 5th, and 95th percentiles obtained from an offline Monte Carlo analysis on the final population with sample size of 500.

[44] Analyzing the results, it is seen that the Pareto front as generated from the final generation of NSGA-II averaged over 5 samples is actually very close to the 5th percentile and no where close to the true average obtained from offline analysis. Because of the small sample sizes used during the GA, solutions with biased sampling in the lower portions of their distributions tend to survive and take over the populations. This effect is especially exacerbated because of the elitist selection scheme being used by the NSGA-II, which causes solutions with low objective values because of biased sampling to always dominate other solutions (which may have more reliable estimates closer to true means), thus

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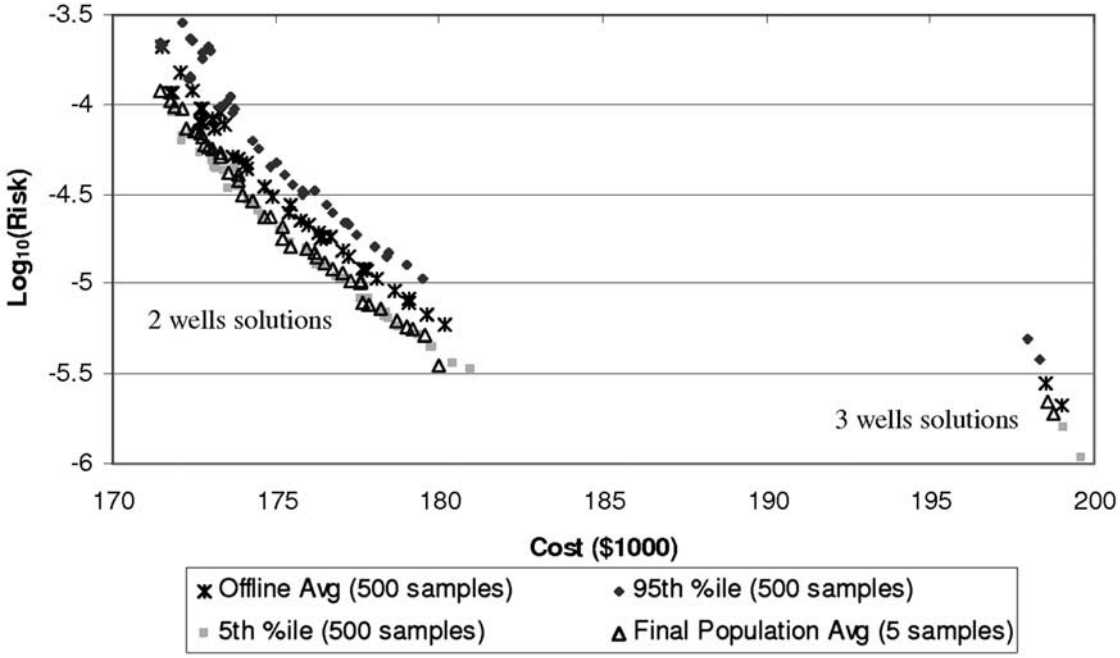


Figure 8. Result of the averaging-based NSGA-II for hypothetical aquifer (showing solutions with low risk/high costs).

surviving into future generations. *Singh et al.* [2003] have shown that removing elitist selection led to sparse and unstable Pareto fronts, so removing elitism was not a good solution to this problem. This bias is seen to persist despite extended sampling because not all solutions benefit from the archiving. The Pareto dominant solutions that appear in the first few generations are sampled again and again as they survive in later generations (because of the elitist selection procedure these solutions always survive). Thus the effective sample size for these solutions increases with time. However, solutions that appear in later generations are not sampled as much and can still suffer from biased selection. In fact, most of the low-risk, high-uncertainty solutions are found later in the runs, so they do not gain from the extended sampling approach and are still far from the expected performance as seen in Figure 8.

[45] As a comparison, the results for the PMOGA (also with extended sampling) on the hypothetical case are shown in Figure 9 (here again the high-cost, low-risk solutions have been shown as in Figure 8). When compared to the previous best results for the averaging approach (Figure 8), it can be seen that the final population (shown by the triangles) has a predicted performance much closer to the actual average (shown by the crosses). To quantify this observation, we compare the root mean square difference between the true average (as given by 500 samples) and the online average (with 5 samples) of the PMOGA with the averaging-based approach. The root mean square difference is calculated by summing the squared difference between the true average and the online average for both objectives (cost and risk) for all of the members of the final population, averaging, and finally taking the square root of these differences. High values of this difference indicate that the cost and risk evaluated online are more biased compared to the true average costs and risks, thus leading to unreliable operations within the GA. For the PMOGA this root mean square difference was found to be 0.098 for cost (in terms of \$1000s) and 0.082 for the (Log_{10}) risk, compared to 0.37 (\$1000 in cost) and 0.27 (Log_{10} risk) with the averaging-based approach. These numbers indicate that compared to the averaging approach the bias for the PMOGA solutions is about 75% less for cost and about 70% less for the risk estimates. Thus, using the probabilistic approach has made the decision making within the GA substantially more robust. In addition, recall that the probabilistic approach explicitly favors solutions with larger effective sample size and lower uncertainty during selection, leading to more reliable results. When the results with this approach are compared with the previous results using simple averaging (Figure 8), there was a decrease of about 10% in the standard deviation for both cost and risk averaged over all the solutions in the Pareto front, as shown in Table 1, indicating that the probabilistic approach discovers solutions with less uncertainty with respect to the conductivity scenarios. Thus in conclusion we can say that the PMOGA finds solutions with less estimation bias as well as lower variability.

[46] For validation of the multiobjective GA, the results from PMOGA were first compared with the results from an averaging-based noisy single-objective genetic algorithm (which has been demonstrated to give reliable solutions with adequate sampling [*Gopalakrishnan et al.*, 2003]) for the hypothetical case with the risk posed as a constraint. The same population and sample size were used for the single objective run. The risk constraint chosen was a (Log_{10}) risk level of -5.5 and solutions were penalized for violating the constraint for any of the realizations (in essence this is similar to the chance constraint approach with a Log_{10} risk constraint of -5.5 requiring 100% reliability on the samples tested during the optimization run). If the PMOGA has converged to the true Pareto front under uncertainty, then this single-objective solution should be part of the trade-off

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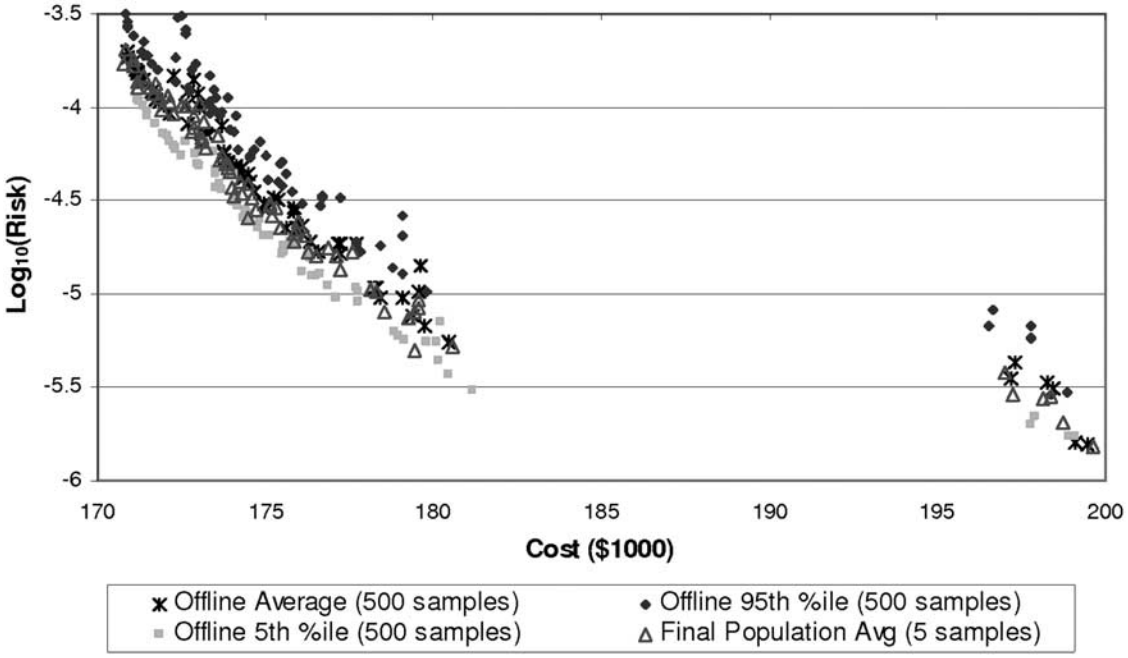


Figure 9. Results of the PMOGA for the hypothetical aquifer (showing solutions with low risk/high costs).

curve for the PMOGA. Figure 10 shows this single-objective solution's mean, 5th, and 95th percentile values from offline Monte Carlo analysis with 500 samples, along with offline average and percentiles (also with 500 samples) of the relevant portion of the PMOGA Pareto front from Figure 9 (costs from $195,000 to $201,000). From Figure 10 it can be seen that the constrained single-objective solution lies on the Pareto band of the multiobjective case. This indicates that the PMOGA has indeed converged to Pareto optimal solutions.

[47] Note also that in this case there is an important discontinuity in the costs for (Log_{10}) risk more than -5.5 (see Figure 9) that can only be identified from the trade-off front generated by a multiobjective approach. For a (Log_{10}) risk level of -5.5 to be satisfied with at least 95% reliability, the cost of the optimal solution is $198,864. However, the cost for a (Log_{10}) risk level of -5 from the PMOGA Pareto front is only $179,786, which is approximately 10% less than that for the risk level of -5.5. This type of tradeoff information cannot be readily obtained from single-objective optimization without multiple trial-and-error runs.

[48] With the offline Monte Carlo analysis, it is also possible to evaluate the costs for different levels of risk and reliability, which is shown in Figure 11. For example, to find the solution with 95% reliability for a given risk level, say (Log_{10}) risk of -5, we simply select the solution with the lowest average cost such that the 95th percentile (from the offline Monte Carlo analysis) is less than (Log_{10}) risk of -5. We can do this for different risk levels as we step through the Pareto front and identify the relationship between cost and risk for a 95% reliability of achieving a particular risk level. This information is shown in Figure 11 as the line labeled "95% reliability." We can also change the required reliability to 99% and repeat the process, stepping through the Pareto front and choosing the solution with lowest cost and the 99th percentile less than the desired risk level - thus generating the line labeled "99% reliability" in Figure 11. This, again, illustrates the type of additional information that can be gleaned from the results of the PMOGA without any additional runs. Analyzing Figure 11, it can be seen that there is a jump in costs when the (Log_{10}) risk levels increase from -3 to -3.5 and from -5 to -5.5. These jumps in cost are due to additional pumping wells that need to be installed to ensure lower risk levels, which offer clear opportunities for savings if the regulator is willing to accept a slightly higher risk level. Moreover if we increase the reliability from 95% to 99%, the cost for a (Log_{10}) risk level of -5 increases such that it is almost equal to the cost of a (Log_{10}) risk level of -5.5 with 95% reliability. Thus uncertainty-based multiobjective optimization allows the stakeholders to analyze trade-offs among all objectives in the face of uncertainty and take appropriate decisions bearing all the criteria in mind. In this case, for example, one could argue that the best decision would be to select the solution that gives a (Log_{10}) risk level of -5 with 95% reliability.

4.2. Results for the Umatilla Application

[49] In this section, we apply the PMOGA algorithm to the Umatilla case to identify the effects of considering uncertainty with multiple objectives at the field scale. The final Pareto front for the Umatilla case is shown in

Table 1. Comparison of Standard Deviations for Both Cost and Risk Found With PMOGA and Simple Averaging

	Cost, $1000	(Log_{10}) Risk
PMOGA	0.156833	0.154632
Simple averaging	0.174623	0.1716219
Percent difference	10.19	9.90

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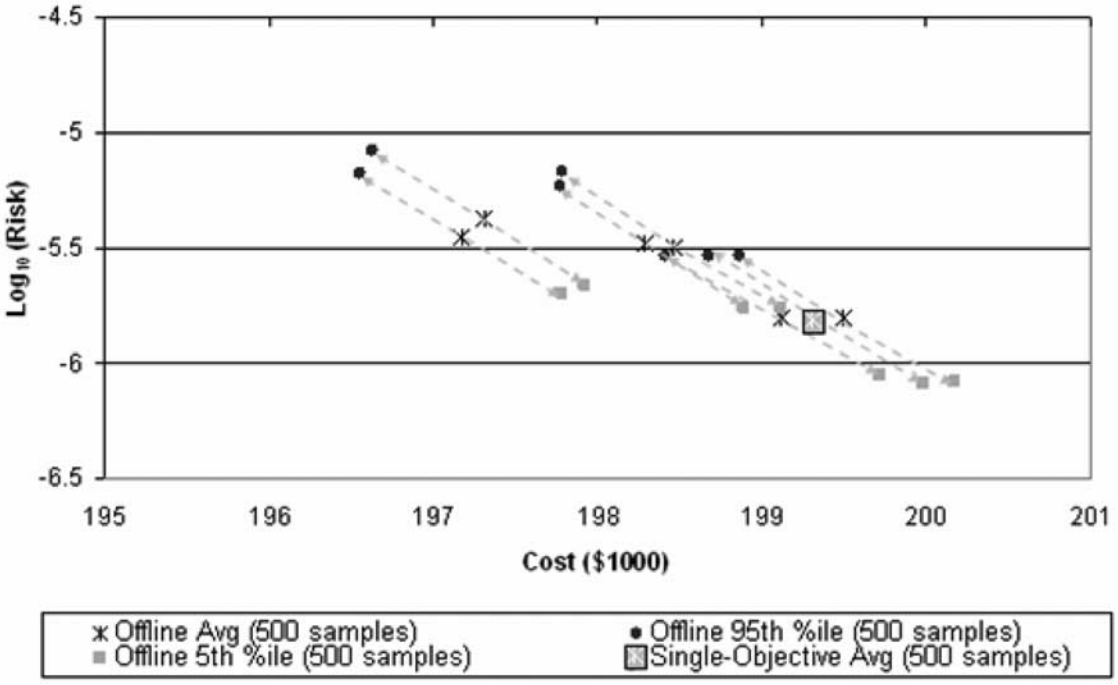


Figure 10. Result of constrained single-objective optimization (with risk level of $10_{-5.5}$) compared to the PMOGA Pareto front for the hypothetical aquifer.

Figure 12a. The open circles in Figure 12a show the average total mass remaining above the cleanup limit for a given remediation cost, as predicted by the GA using 5 new samples in the final generation (of course, extended sampling increases the effective sample size for these solutions). As before, these solutions were then evaluated using off-line Monte Carlo simulation with all of the 96 realizations to better estimate the "true" behavior of each solution over all conductivity realizations. The average, 5th, and 95th percentiles from the Monte Carlo simulation are shown by the diamonds, triangles, and squares respectively. The solution marked "no cleanup" corresponds to the minimum cost (corresponding to no pumping at existing or new wells or injection basins, but only sampling and monitoring costs) and the maximum mass remaining in the system. For this case, there is no uncertainty in the mass remaining with

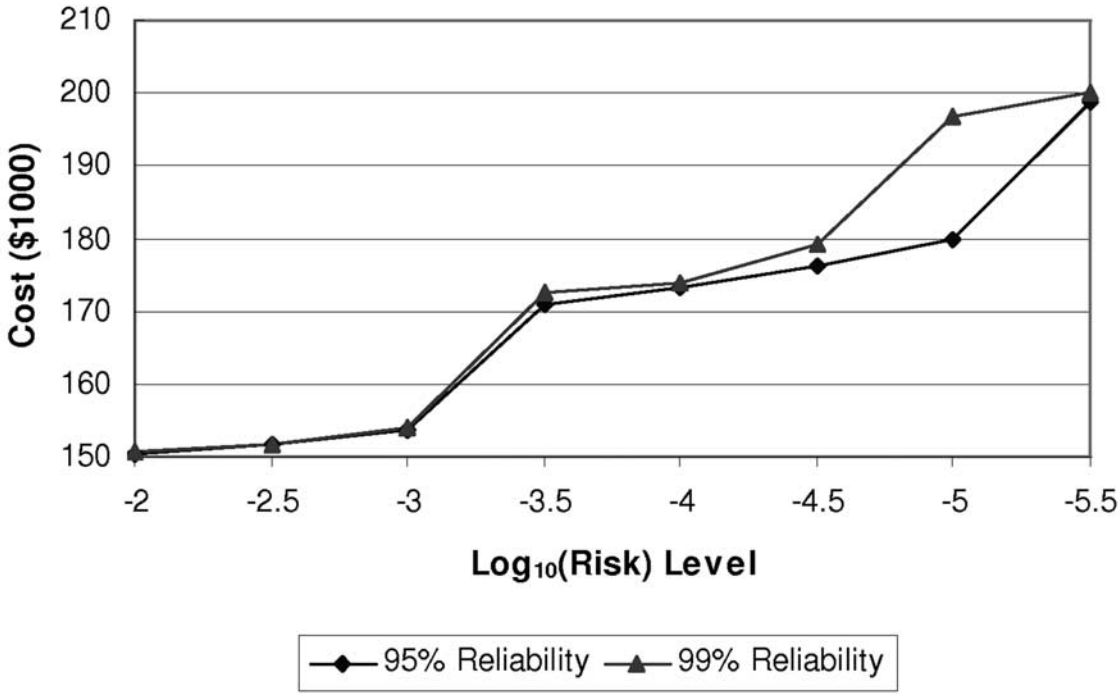


Figure 11. Risk levels and associated costs for two reliability levels.

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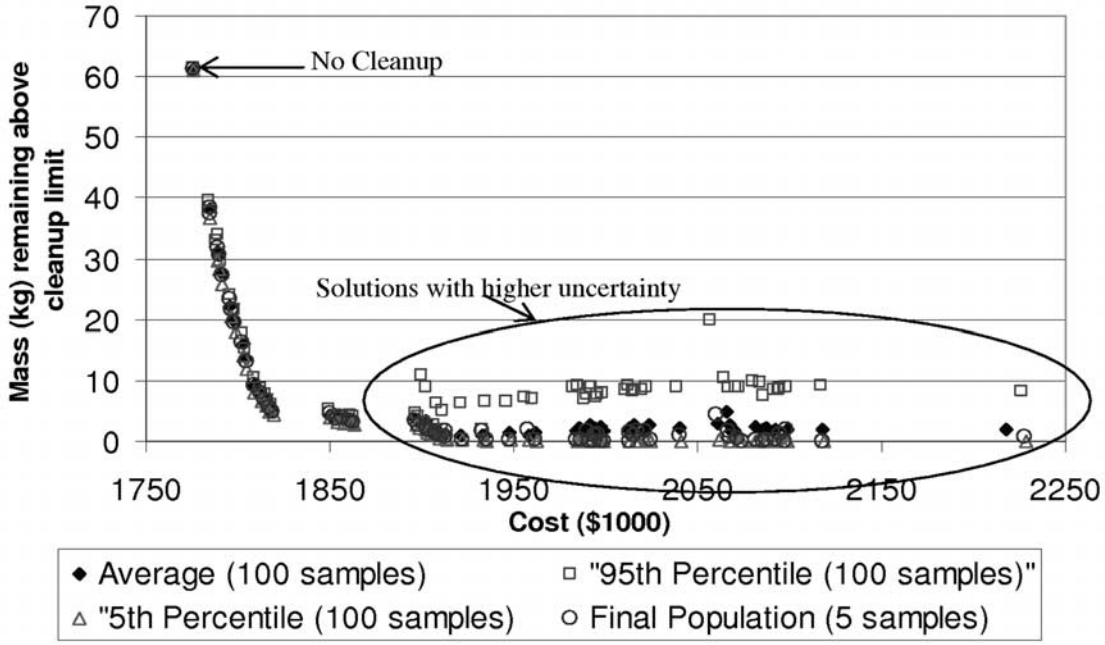


Figure 12a. Average, 5th, and 95th percentiles for the cost-cleanup tradeoff for the Umatilla Chemical Depot remediation solutions.

respect to the hydraulic conductivities, as seen by the convergence of the mean, 5th, and 95th percentiles for this solution. This is because there is no pumping and the total mass remaining of this conservative contaminant is simply the initial mass in the aquifer (which is treated as deterministic in this case).

[50] To better visualize the remaining results that had uncertainty, the tradeoff curve is shown only for the low mass-remaining levels in Figures 12b and 13. Figures 12a and 12b show that there is a significant tradeoff between the cost and the cleanup efficiency, but as the cost increases the relative improvement in cleanup efficiency decreases substantially. In general, higher costs mean more intensive pumping or a larger number of wells, and as can be seen there is much greater uncertainty for the more expensive solutions with lower mass remaining. One of the reasons for higher uncertainty in the low-mass solutions is that, as the mass is cleaned up from the site, localized islands of high concentrations remain that correspond to areas with low permeability. The location and amount of this residual mass

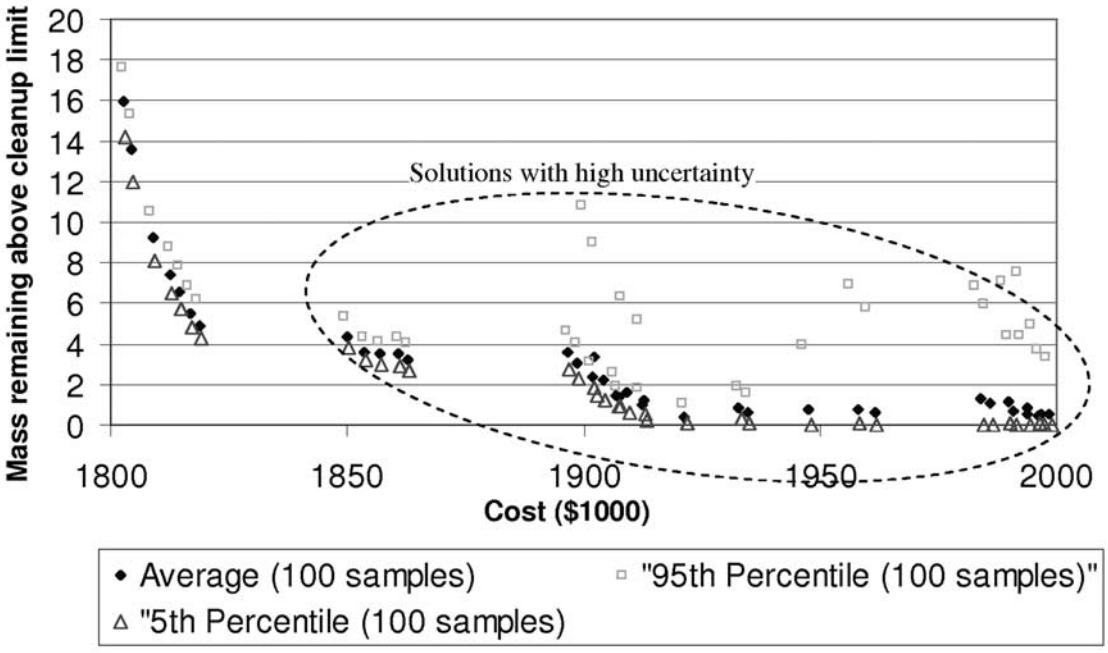


Figure 12b. PMOGA solutions with low mass remaining (and high uncertainty).

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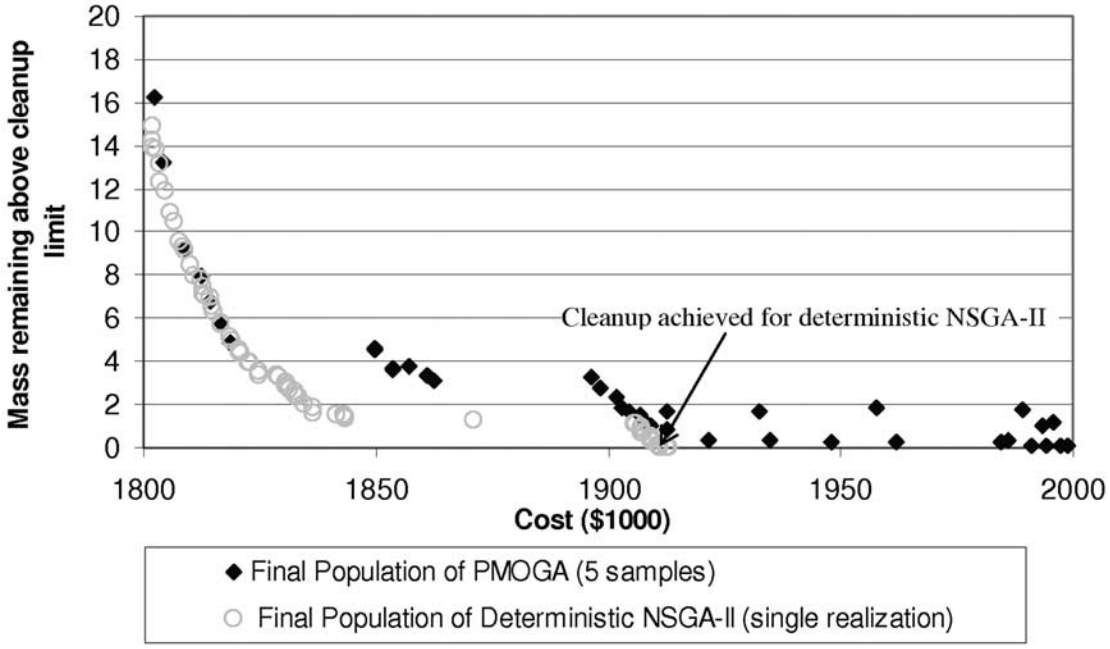


Figure 13. PMOGA Pareto front compared with the deterministic Pareto front (for low mass-remaining solution).

depends heavily on the hydraulic conductivities, producing greater variation when these solutions are sampled over the different realizations. This is one reason why the probabilistic decision making approach of the PMOGA, which takes into account differences in mean and the variance of the solutions, is important for this problem. Applying the simple averaging approach would neglect the effect of the differences in variance between different cleanup solutions. It is possible that this may be a characteristic of remediation systems under uncertainty, but has not been observed previously because other studies have not examined trade-offs under uncertainty. Further research needs to be conducted to see if this is indeed true for other field-scale applications. It is noteworthy that no solution gave cleanup for all the realizations within the 5-year remediation period simulated (note how the 95th percentile for the mass remaining above cleanup limit does not reach zero for any of the solutions). Thus most solutions require longer than 5 years to clean up the aquifer reliably. Recall that 5 years was the optimal cleanup length found in the ESTCP project when uncertainty was not considered. An important future research question would be to examine the effect longer remediation periods would have on the results.

[51] It is also important to investigate what difference considering uncertainty makes to the Pareto front and solution quality. To do this, results from the PMOGA are compared with results from a deterministic version of the same algorithm, which uses only the original hydraulic conductivity realization selected by the Army Corps of Engineers. Figure 13 shows the Pareto optimal front found by the deterministic NSGA-II (shown by the light circles) compared to the Pareto optimal front generated by the PMOGA (the offline analysis for the PMOGA are not shown to improve the legibility of the graph). The deterministic NSGA-II was run with exactly the same population, number of generations, and other GA parameters as the PMOGA but only one conductivity scenario was used to estimate the objective function values (the same scenario used by USACE and *Minsker et al.* [2003]). The deterministic solution is the same as the optimal solution found by *Zheng and Wang* [2002] and *Peralta* [2002] for the ESTCP project (their results are summarized in a forthcoming paper by *Becker et al.* [2006]). Note that the deterministic NSGA-II does not have a long tail with high costs and low mass. As explained previously the long tail in the PMOGA front is due to uncertainty of solutions with low mass remaining. For these solutions, it is not possible to statistically decide among any of the solutions with the given significance level, so they are allowed to remain in the final nondominated front, providing multiple solutions with low mass remaining and varying levels of uncertainty and cost. The deterministic NSGA-II finds a solution where cleanup is achieved in 5 years for the given conductivity scenario. To test the uncertainty of this solution, we conducted offline Monte Carlo analysis, and calculated the average costs, mass remaining, and standard deviation for this solution. The solution for the deterministic NSGA-II was found to have an average cost of $1,913,000, an average mass of 0.89 kg of contaminant over the cleanup limit, and a standard deviation of 1.76 kg (for the mass remaining). Thus, even though this solution predicts cleanup for the one conductivity scenario, it does not achieve cleanup for other conductivity scenarios. As can be seen from Figure 13 there are solutions on the PMOGA front with similar cost and mass remaining to this deterministic solution. The PMOGA solution closest to the deterministic case (in terms of cost and mass remaining) has a cost of $1,912,000, an average mass of 0.98 kg, and standard deviation of 0.38 kg (for the mass remaining). The cost and average mass remaining for these two solutions are

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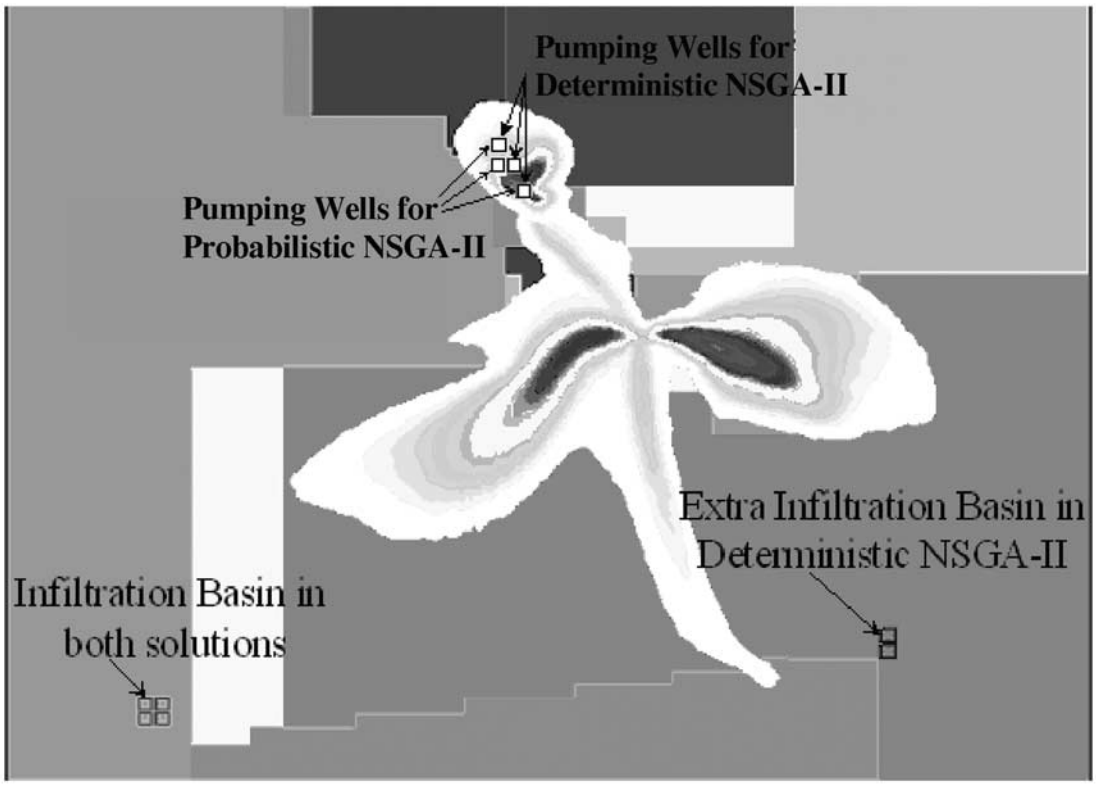


Figure 14. Comparison of well configurations for a deterministic NSGA-II solution with a PMOGA solution with similar cost and average mass.

essentially the same, the small difference being due to the fact that the pumping wells for the deterministic case have a slightly higher pumping rate (leading to slightly higher cost and lower average mass remaining). However, the estimated standard deviation in mass remaining for the deterministic solution was nearly five times the standard deviation of the PMOGA solution with similar cost and cleanup efficiency. This again indicates that the probabilistic NSGA-II finds more reliable Pareto optimal solutions. As an aside we would like to note that the PMOGA solution chosen here is also the solution with the lowest standard deviation among the solutions with low mass remaining.

[52] To investigate why the PMOGA solution results in lower standard deviation of the mass remaining we show the actual well configurations for both the deterministic and PMOGA solutions in Figure 14. Figure 14 shows that the two solutions are very similar except for the position of one extraction well and the addition of one extra infiltration basin to the deterministic solution. To find out which of these two changes contributed more to the variability, the extra injection basin was removed and the resulting solution was evaluated using all realizations. The same process was repeated by changing the position of the different extraction well. It was found that the position of the extraction well had the most significant impact on the variability of the mass remaining with respect to different conductivity scenarios. The deterministic NSGA-II was finding a solution with the well installed in a region with lower saturated thickness. The PMOGA, on the other hand, installed the extraction well in a deeper portion of the aquifer with more saturated thickness. This leads to more reliable pumping as the extra thickness allows for a higher effective transmissivity for the same conductivity values, thus making the solution less dependent on the conductivity of the aquifer around the well. This is consistent with a trial-and-error sensitivity analysis carried out by *Peralta and Kalwij* [2003] in the ESTCP project, which found that installing wells along a northwest-southeast depression in bottom elevations of the aquifer allowed for more robust pumping at the site. It is interesting to note that the PMOGA discovered this site characteristic and installed the extraction well in that very zone of extra thickness.

[53] Finally, note that unlike the earlier example where it was possible to select a risk level and assess the reliability of the solution, here it is not reasonable to do so because a majority of the realizations still have some mass remaining. In such a situation it is desirable to select the solution with the lowest cost, mass remaining, and uncertainty. However, there is a clear trade-off among these three criteria. Figure 15 plots the standard deviation (as evaluated from offline analysis with all the realizations) of mass remaining for the PMOGA solutions with low (less than 1.25 kg) average (offline) mass remaining. Figure 15 also shows the standard deviation and average mass remaining for the "deterministic" solution as well as for the "chosen" PMOGA solution (discussed in the previous paragraphs). From Figure 15, it can be seen that this chosen PMOGA solution compared earlier to the deterministic case is actually the case with the lowest standard deviation. However, there are other solutions with lower average mass remaining but higher uncer-

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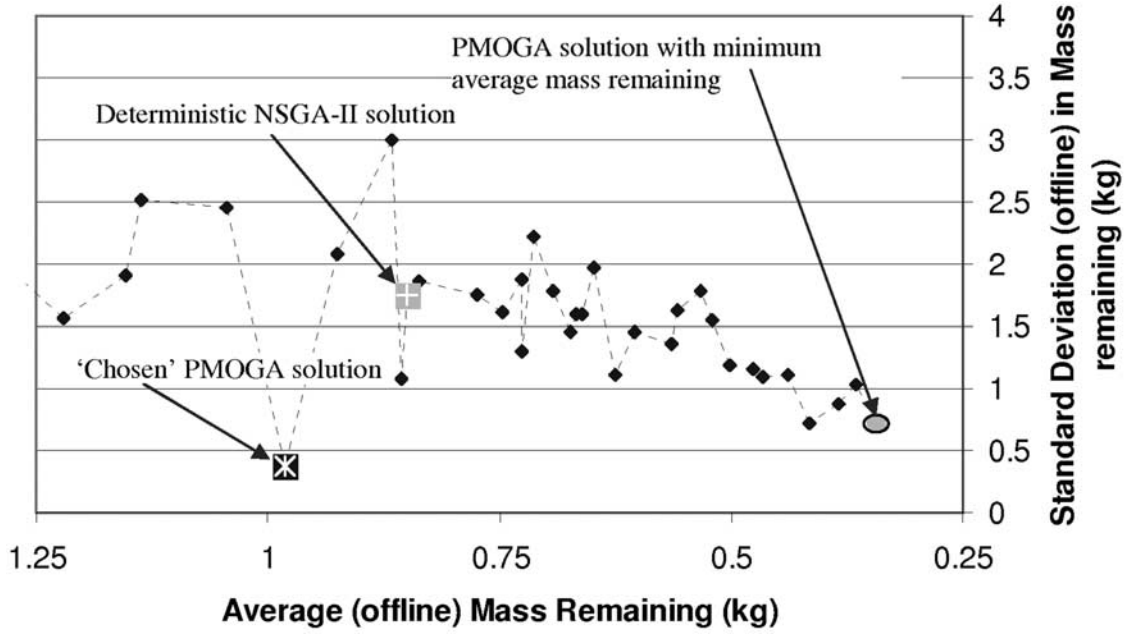


Figure 15. Mass remaining and standard deviation (uncertainty) for low mass-remaining PMOGA solutions.

tainty. For example, the solution corresponding to the lowest average mass remaining is also indicated in Figure 15. This solution has a cost of $1,921,000, average mass remaining of 0.72 kg, and standard deviation of 0.41 kg. If the average mass remaining is the overriding criteria for the decision maker then he or she can select this remediation solution, bearing in mind that it is more expensive (by about $9000; the chosen PMOGA solution has a cost of $1,912,000) and more dependent on the aquifer conductivities (since it has higher standard deviation, 0.41 kg compared to 0.38 kg, with respect to the conductivity realizations) compared to the chosen PMOGA solution with the lowest standard deviation.

5. Conclusions

[54] This work investigates the simultaneous effects of uncertainty on multiple objectives for groundwater remediation design using a framework called PMOGA that combines a multiobjective genetic algorithm with a probabilistic ranking and distance calculation to improve decision making under uncertainty. This approach was developed and tested on a hypothetical case study where significant improvements were demonstrated over a more simplistic "averaging" approach. It was demonstrated that the probabilistic multiobjective approach allows the user to consider the entire trade-off among the various objectives in the face of uncertainty, which was not previously possible.

[55] The PMOGA was then implemented on a field-scale case study at the Umatilla Chemical Depot in Hermiston Oregon, which was the subject of a recent transport optimization demonstration. The results demonstrated that including uncertainty and multiple objectives can give significant information with respect to the cleanup design. The high-cost and low-residual mass solutions were found to have substantially more uncertainty at this site. Thus as cost was increased the reliability was seen to decrease, indicating that more intensive and costly remediation strategies might have lower reliability even though they have better efficiencies. Furthermore, it was seen that the 5-year cleanup period was not sufficient to guarantee cleanup over all considered conductivity scenarios for this case study. This case illustrates a situation where the mass remaining needs to be treated as a minimization objective and a constraint-based optimization approach would not be appropriate. Comparing a solution found by the probabilistic method with the optimal deterministic solution, the new approach found a small change in the design that yields a more reliable solution with very little increase in cost. The multiobjective approach can facilitate this type of decision by giving multiple solutions to the decision maker with different levels of cost, mass remaining, and uncertainty. With such information the decision maker can then evaluate different solutions and select the solution with the best compromise between reliability and cleanup efficiency.

[56] The results also demonstrated the importance of uncertainty for different remediation design problems. It was seen that parameter uncertainty has a very significant effect on the effectiveness of intensive remediation strategies, especially for the Umatilla case study. Thus, if the goal is to clean up the site reliably in a short time period, it is imperative in this case to invest in data collection and better site characterization. This information can be used not only to reduce the uncertainty in the remediation strategy chosen, but also to make a more informed and reliable decision when choosing a remediation strategy from the final trade-off front.

[57] These findings demonstrate the advantages of addressing uncertainty in multiple criteria within a Pareto-optimization framework. The ranking and crowding schemes proposed in this work can easily be used in other multiobjective optimization frameworks [e.g., *Corne et al.*, 2000; *Alberto et al.*, 2003; *Van Veldhuizen and Lamont*,

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2000] when considering uncertainty. The most significant advantage to using such a population-based approach is that it allows the user to simultaneously identify the most likely Pareto-optimal solutions along the "Pareto region." In addition, using the probabilistic crowding scheme allows the optimization algorithm to favor solutions with lower uncertainty while maintaining diversity across the Pareto region. It is important to note that such an approach is advantageous only when the decision maker is interested in evaluating the trade-off among multiple (uncertain) objectives. If it is sufficient to address uncertainty only in the constraints while considering a single optimization objective, then chance-constraint [*Charnes and Cooper*, 1959] or related approaches [*Morgan et al.*, 1993; *Chan*, 1994; *Guan and Aral*, 2004] may be more appropriate. If uncertainty need only be considered in a single objective, then noisy single-objective approaches such as *Miller* [1997], *Gopalakrishnan et al.* [2003], or *Beckford et al.* [2003] would be more appropriate.

[58] One limitation of using a population-based optimization approach such as the NSGA-II is the computational burden imposed by fitness evaluations, particularly when optimizing under uncertainty since each solution has to be evaluated multiple times to obtain the mean and standard deviation for each objective. Since small sample sizes need to be used within the GA, postoptimization uncertainty analysis also needs to be undertaken to rigorously assess the solution quality. To reduce some of this computational burden the concept of extended sampling (using an archive of past evaluations) was used in this work. Over the past few years many promising approaches have been proposed to reduce the computational burden of population-based approaches such as genetic algorithms. These include parallel and "multiscale" approaches that split the computation of the objective function among multiple-variable grid processors [e.g., *Eby et al.*, 1997; *Lin et al.*, 1997; *Cantú-Paz*, 2000; *Babbar and Minsker*, 2003]; hybrid approaches that combine the global search of a GA with more local gradient-based search algorithms, [e.g., *Hsiao and Chang*, 2002; *Kapelan et al.*, 2002; *Espinoza and Minsker*, 2006], and fitness approximation approaches that build fast surrogate models to replace the fitness function calculation of the GA [e.g., *Brooker et al.*, 1998; *Jin et al.*, 2002; *Peralta*, 2002]. In the future, research done in these fields will be investigated to incorporate into the PMOGA framework to further reduce the computational requirements of such an approach.

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